

Multiple Choice

1. **Answer:** D

Options: A) Snort; B) SuperScan; C) Burp Suit; D) EtterPeak

Note: EtterPeak is recognized as a popular tool for network analysis in multiprotocol diverse networks due to its ability to analyze and visualize network traffic across various protocols effectively, making it essential for understanding complex network behaviors.

2. **Answer:** B

Options: A) 160; B) 128; C) 64; D) 54

Note: The correct answer is 128 because MD 5 (Message Digest Algorithm 5) consistently produces a fixed-length output of 128 bits, regardless of the input message size.

3. **Answer:** A

Options: A) Receiver; B) Sender; C) Modulator; D) Translator

Note: The correct answer is "Receiver" because message confidentiality ensures that only the intended recipient can comprehend the content of the message, protecting it from unauthorized access and ensuring privacy during transmission.

4. **Answer:** A

Options: A) 160 bits; B) 512 bits; C) 628 bits; D) 820 bits

Note: SHA-1 produces a fixed-length output known as a message digest that is 160 bits long, making this the correct answer.

5. **Answer:** A

Options: A) Silk Road; B) Cotton Road; C) Dark Road; D) Drug Road

Note: The Silk Road was a notorious Dark Web marketplace that specialized in the illegal trade of drugs and various other illicit goods, making it a prominent example of such platforms.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: The statement is false because EXE does not restart the STP solver upon a timeout; instead, it typically employs a different strategy for handling timeouts, such as returning an unsatisfiable result rather than retrying the query.

7. **Answer:** False

Options: A) True; B) False

Note: The statement is false because scanning types primarily refer to methods used in

information security and network vulnerability assessments, rather than being specific to software testing.

8. **Answer:** False

Options: A) True; B) False

Note: The Heartbleed bug exploits a vulnerability in the OpenSSL library's heartbeat extension that allows attackers to read sensitive memory, but it does not facilitate a format string attack, which is a different type of vulnerability involving improper handling of string formatting in programming.

9. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the one-time pad (OTP) key must be a truly random key that is as long as the message, and computing the key as $k = m \text{ xor } m$ results in a key of all zeros, which does not provide the necessary randomness for secure encryption.

10. **Answer:** True

Options: A) True; B) False

Note: Penetration testing involves simulating cyber attacks on a system to comprehensively evaluate its security posture, thereby identifying vulnerabilities and weaknesses throughout the entire system.

Long Form Response

11. **Answer:** `def count_list(input_list): | return (len(input_list))**2`

Note: The provided function correctly counts the number of sublists in the given list of lists using the 'len()' function and then squares that count, fulfilling the requirements of the question.

12. **Answer:** `def neg_nos(list 1): | return num`

Note: correct because it defines a Python function that is intended to filter and return negative numbers from a provided list, demonstrating the basic principles of function definition and list manipulation in Python.

End of Answer Key